Trade Policy Sentiment and Corporate Investment

Dario Caldara Francesco Ferrante Matteo Iacoviello Immo Schott

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Motivation

- Beginning in early 2025, the U.S. imposed substantial new tariffs; the average effective rate reached levels not seen since the 1940s
- Estimating the effects of these policies is challenging:
  - Measures are heterogeneous across countries and products
  - Multiple transmission channels
  - Macroeconomic conditions confound simple comparisons
- Earnings calls provide timely information about firms' tariff assessments and investment plans, potentially helping explain subsequent investment.
- Connecting firm-level evidence to a macroeconomic model allows us to assess aggregate investment effects.

What we do

1. **Identify tariff mentions** in earnings call transcripts.
2. **Score tariff sentiment** in the extracted text using an LLM.
3. **Link to Compustat** firm-level data.
4. **Analyze the correlation with investment** using regressions and synthetic controls.
5. **Quantify aggregate effects.** Use a two-sector open economy model to recover the response missing from relative comparisons.

$\Rightarrow$ Aggregate investment was **3.5% lower** in 2025 than without the tariff increases.

Earnings calls data

- Earnings calls are quarterly calls in which executives discuss results and the outlook, then answer analysts' questions. Transcripts record these exchanges verbatim.
- **Source:** S&P Global earnings call transcripts for U.S. firms, 2024Q1–2026Q2.
- Find "tariff" and "tariffs"; retain 50 word snippets on each side.
- Score each snippet, using Claude Opus 5.5.
- Distinguish texts without tariff mentions from neutral discussion.

Scoring tariff sentiment

Analyze this company's sentiment toward trade tariffs using a -1 *to* $+1$ *scale, in increments of* 0.1.

-1.0	Major threat or cost
-0.5	Some concerns
0.0	Neutral, unclear or balanced
$+0.5$	Some benefits
$+1.0$	Major opportunity or advantage

- Extensive human validation: manual review of tariff passages and tests on constructed examples spanning the sentiment scale.
- Research assistants scored 193 randomly sampled excerpts using the same prompt and scale as the LLM. Human scores Dictionary

Tariffs can be an opportunity or a threat

Opportunity: Cleveland-Cliffs, February 25, 2025

"We at Cleveland-Cliffs appreciate the recently announced 25% tariffs on steel imports from all countries."

Threat: Apple, May 1, 2025

Expected tariff costs in the June quarter, assuming unchanged tariff policy:

"[W]e estimate the impact to add $900 million to our costs."

Tariff assessments and investment decisions

Wynn Resorts, May 6, 2025

"[T]he current tariff rates have driven us to delay about $375 million of CapEx projects, including the Encore Tower remodel."

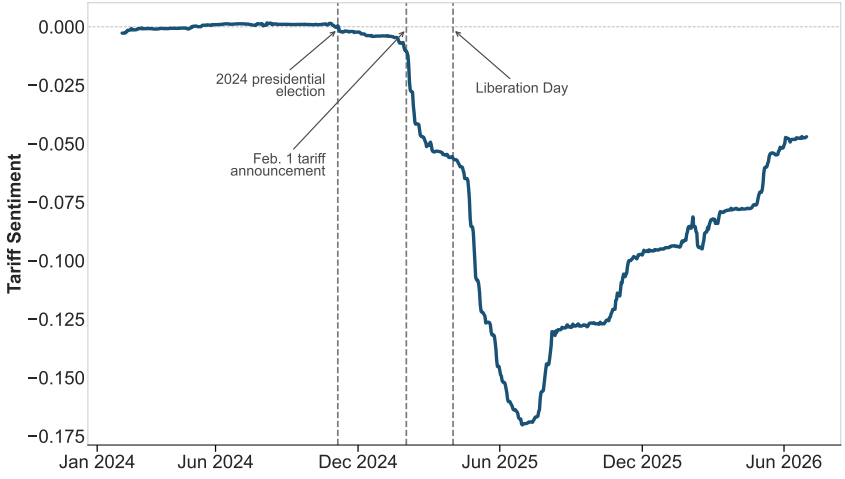
General Motors, January 28, 2025

"What we won't do is spend a large amount of capital without clarity."

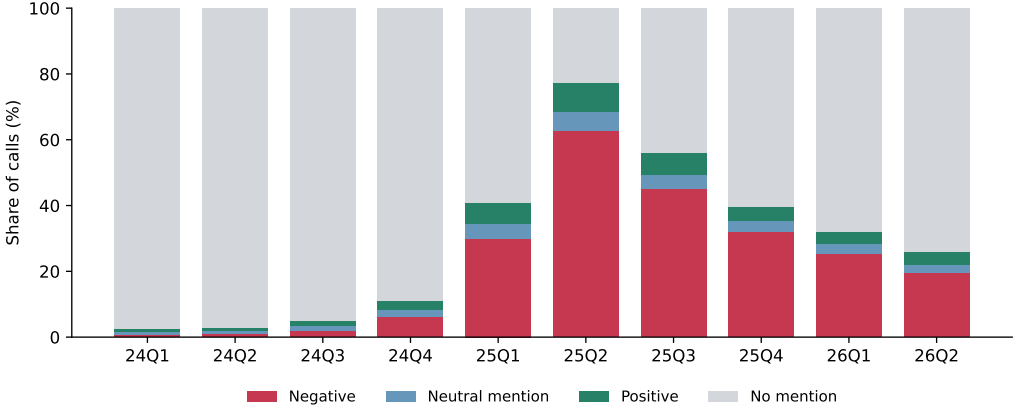
Nucor, July 29, 2025

"Tariff policy continues to evolve, but has been positive for the steel industry overall."

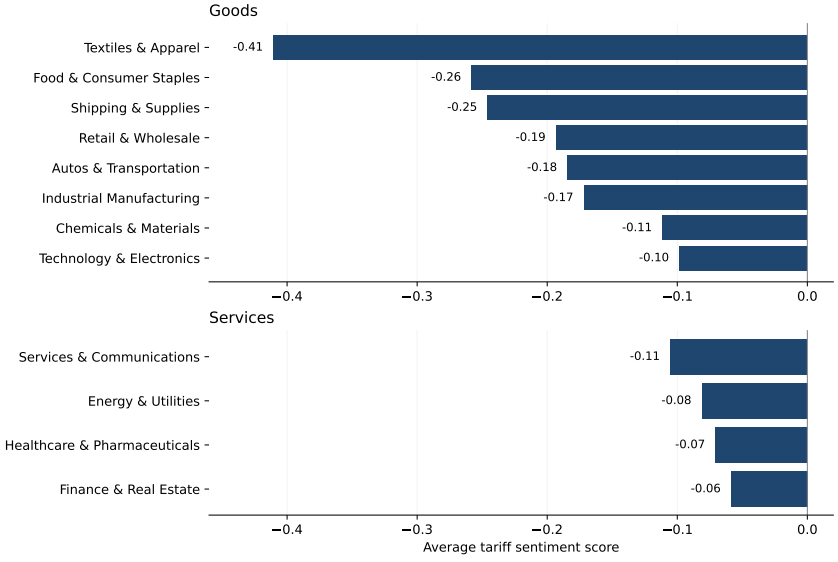
Average tariff sentiment since 2024



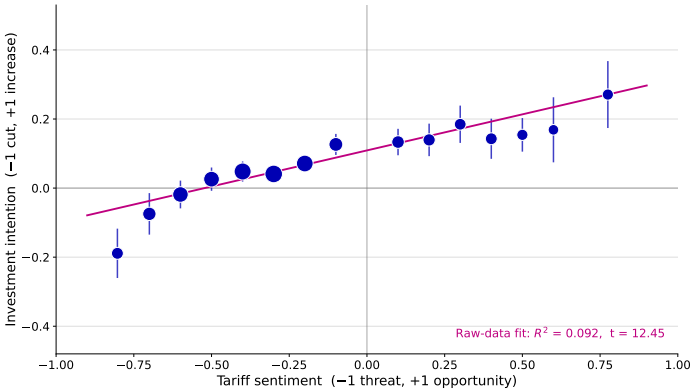
Heterogeneity in tariff sentiment



Industry heterogeneity (2025 avg.): trade exposure matters



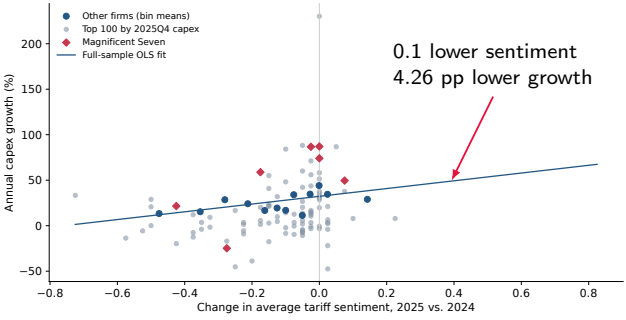
Sentiment and stated investment plans



- A separate prompt scores tariff-induced investment plans in the same call.
- Positive sentiment: 93% report positive plans.
- Negative sentiment: 39% report cuts, but 61% still plan to invest, including adaptation.

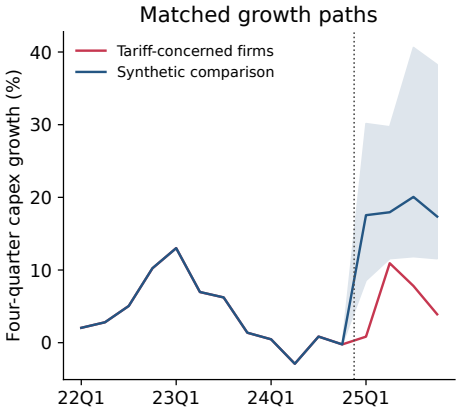
Concern: could firms blame tariffs for investment weakness caused by other factors? Stock-price check

Lower sentiment is associated with weaker investment growth



- Average sentiment fell by 0.116 points from 2024 to 2025, implying 4.94 pp lower firm investment growth. Regressions
- Robust to AI sentiment controls, excluding the Magnificent Seven, and asset weighting. Panel regs
- U.S. private nonresidential fixed investment: \$4.091tn (2024), \$4.366tn (2025); +5.4% real.

Synthetic control: Investment growth



- **SC:** match tariff-concerned firms' pre-2025 growth with a comparison portfolio.
- **NIPA weights:** fit pooled growth through 2024 to track NIPA private nonresidential fixed investment growth 2022-2024.
- **Result:** average 2025 growth gap of -12.36 pp.

Placebo gaps Before matching

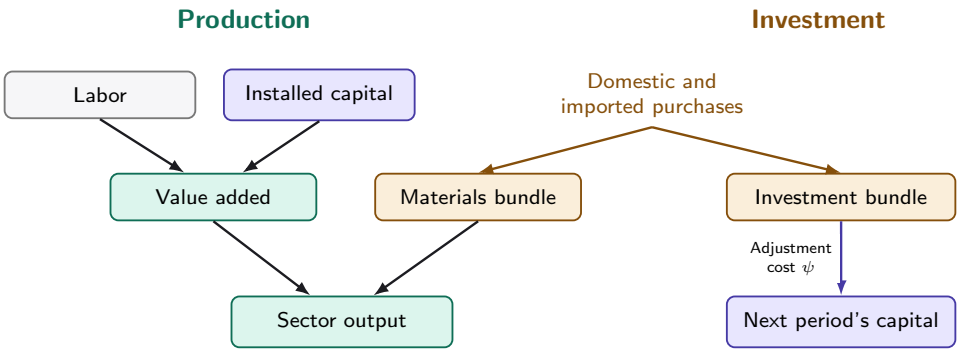
Model

- Two-country open-economy trade model with goods and services sectors, intermediate inputs, and capital.
- Why a model? Zero tariff sentiment does not imply zero tariff effects: GE effects through prices, monetary policy, &c. also affect comparison firms.
- Use the relative investment evidence to quantify aggregate effects, including the response of comparison firms.

High-level model overview

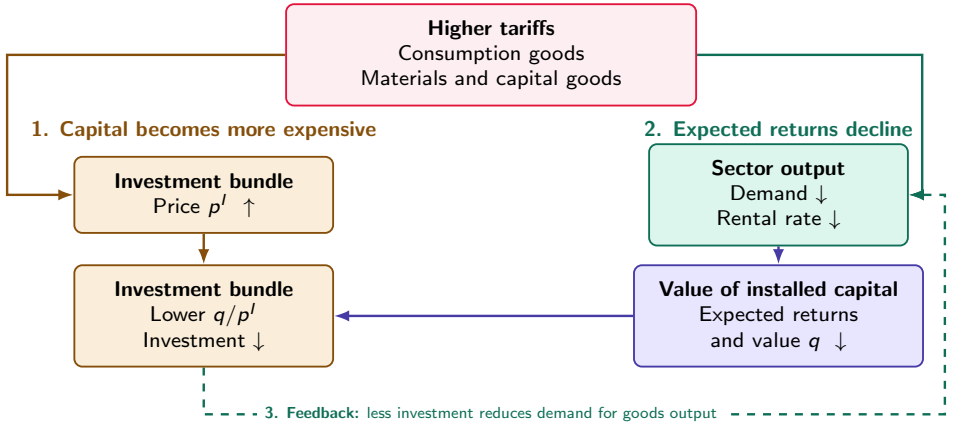


Production and investment



Goods firms import 25% of their materials and investment bundles; services firms import 1%.

Experiment: An increase in tariffs



Aggregate effects of the tariffs



Full IRFs | IRFs with decomposition

Conclusion

- **Measurement:** Assess tariff sentiment using firm's earnings calls.
- **Link with firms' balance sheets:** More negative sentiment correlates with weaker investment.
- **Theoretical model:** Translate relative investment responses into aggregate effects.
- **Results:** Tariffs had a sizeable negative effect on aggregate investment.

Appendix

Details: tariff experiment

- U.S. import tariffs apply to consumption, intermediate, and investment goods. Pre-shock tariff rate: 2.5%.
- Tariff persistence: $\rho_\tau = 0.9$ per quarter. News arrives one quarter before the anticipated change.

2025 quarter	Q1	Q2	Q3	Q4
Realized tariff rate (%)	2.5	8.0	10.7	11.25
News for next quarter (pp)	5.0	5.0	2.5	0.0

Realized innovations reconcile earlier announcements with the observed tariff path; announced increases do not all materialize as expected. The same tariff path applies to all three import uses.

- Goods firms import 25% of materials and investment bundles; services firms import 1%.
- Expectations and costs of changing investment growth allow investment to respond before tariffs take effect.

◂ Back to mechanism Calibration

Two investment responses discipline the calibration

- **Tariff target:** 7.5 pp first-year goods–services investment-growth gap in the calibration; especially informative about ψ.
- **Monetary target:** 2% average first-year aggregate investment decline after a 100 bp monetary shock; informative about κ and ψ jointly.

Parameter	Role	Value
$\kappa = \kappa_w$	Price and wage adjustment costs	110
ψ	Investment-growth adjustment costs	1.1
$\eta_c = \eta_m = \eta_I$	Domestic/import substitution	1.5
ρ_τ	Quarterly tariff persistence	0.9

Fitted responses: tariff gap -7.52 pp; monetary investment response -2.05%. The 7.5 pp calibration target has not been replaced by the refreshed 12.36 pp firm gap. International-bond portfolio costs are separate from investment adjustment costs.

◂ Back to mechanism Tariff details

Validation against human judgments

Same 193 excerpts	LLM	Dictionary
Correlation with human scores	0.703	0.12
Negative / neutral / positive agreement	76%	35%

- Human coders used the same prompt and scale on 193 random excerpts. 82% of LLM scores are within ± 0.2 of the human score; rank correlation is 0.730.
- Accounting language can describe a tariff cost without negative words. Competitive benefits can appear in language a dictionary calls negative.
- A separate prompt measures tariff-induced investment plans; stock returns provide an external outcome check.

Back to sentiment scoring

Table 2, Panel A: quarterly investment regressions

	(1) Baseline	(2) + AI	(3) Same sample	(4) + Size	(5) + All controls
Tariff sentiment	0.402**	0.400**	0.473***	0.446**	0.447**
	(0.185)	(0.185)	(0.181)	(0.180)	(0.177)
AI sentiment		0.108	0.078	0.087	0.081
		(0.142)	(0.145)	(0.144)	(0.143)
Log assets				0.790**	0.504
				(0.356)	(0.353)
Cash / assets					5.283***
					(0.935)
Debt / assets					-0.898*
					(0.537)
Sales growth (%)					0.00003
					(0.00003)
Firm fixed effects	Yes	Yes	Yes	Yes	Yes
Quarter fixed effects	Yes	Yes	Yes	Yes	Yes
N	20,739	20,739	19,025	19,025	19,025
Firms	2,619	2,619	2,475	2,475	2,475

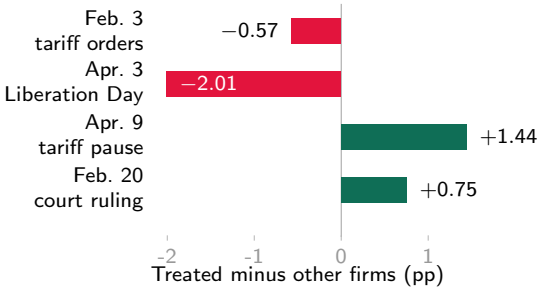
Outcome: $100 I_t / PPE_{t-1}$, 2024Q2–2026Q2. Firm-clustered standard errors; sentiment and controls precede the outcome. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Interpretation: a 0.1-point deterioration in sentiment is associated with a 0.045 pp lower next-quarter investment rate (column 5). The outcome is a quarterly investment rate, not annual capex growth.

◂ Back to cross-sectional regression

External check: stock prices react to tariff news

Back



Day-0 abnormal-return gap, market model.

Caveat: This does not establish causality. The headline classification uses 2025 calls, including information recorded after some events.

The sign follows the tariff news

- Tariff-concerned firms underperform after the February and April tariff increases.
- They outperform after the pause and court ruling.
- With FF49 industry effects and size controls: April 3 is -1.12 pp and the court ruling is $+0.47$ pp.

Interpretation: external validation that sentiment contains firm-specific tariff exposure.

Table 1: Cross-sectional regressions

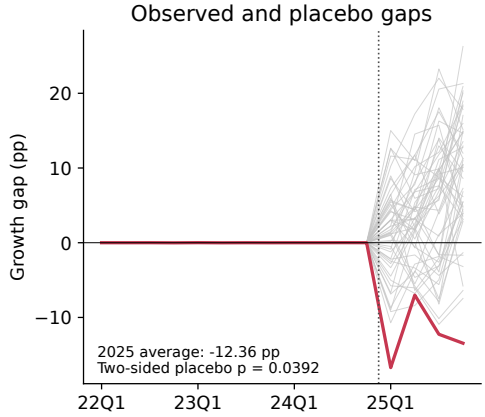
Back

	(1) Baseline	(2) 2025 level	(3) 2024 level	(4) + AI	(5) Asset wtd.	(6) Top 100
Δ tariff sentiment	42.60***			42.94***	34.32***	53.02**
	(14.19)			(14.18)	(5.99)	(21.10)
2025 sentiment level		47.87***				
		(12.62)				
2024 sentiment level			78.77**			
			(31.92)			
Δ AI sentiment				-17.80		
				(12.46)		
N	1,990	2,083	2,115	1,990	1,990	100
Mean growth (%)	27.35	27.84	26.53	27.35	15.02	18.07
R^2	0.005	0.007	0.003	0.006	0.016	0.061
Firm fixed effects	No	No	No	No	No	No

Outcome: annual capex growth, 2025 versus 2024, winsorized at p2.5/p97.5. Δ sentiment = 2025 minus 2024 annual mean; columns (2)–(3) use annual sentiment levels instead. Column (5) uses lagged asset weights, including its mean-growth entry. Standard errors in parentheses.

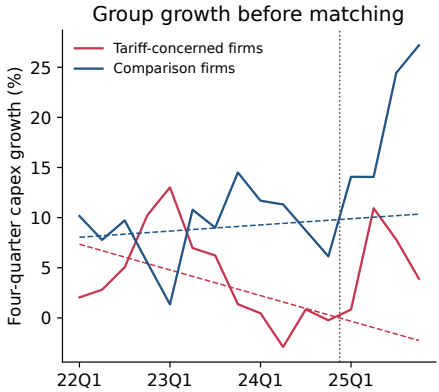
Magnificent Seven exclusion: 42.18 (SE 14.23), $N = 1{,}983$. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Synthetic comparison: observed and placebo gaps



Back to matched paths

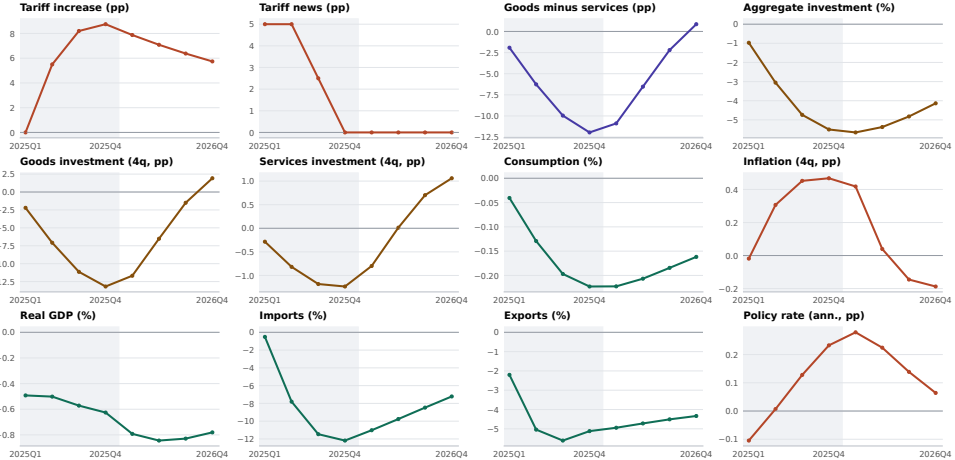
Synthetic comparison: growth before matching



Before matching, the groups already diverge in 2024. NIPA weights fitted through 2024.

Back to matched paths

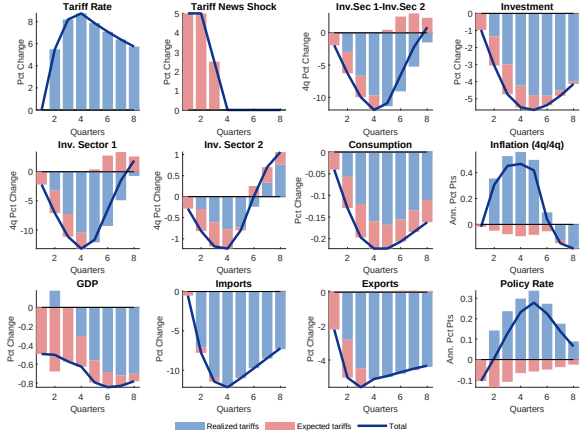
Full model responses



◂ Back to aggregate effects

With decomposition

Full model responses: news and implementation



Back to aggregate effects

Without decomposition